

1 D

$$\text{Resistivity, } \rho = \frac{RA}{l} = \frac{R\pi(d/2)^2}{l}$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta R}{R} + 2\left(\frac{\Delta d}{d}\right) + \frac{\Delta l}{l} = \frac{1}{68} + 2\left(\frac{0.1}{5.0}\right) + \frac{1}{25}$$

$$\Delta\rho = 0.5 \times 10^{-2} \text{ } \Omega \text{ m (1 s.f.)}$$

$$\therefore \rho = (5.3 \pm 0.5) \times 10^{-2} \text{ } \Omega \text{ m}$$

2 C